Topic 1 Design of Experiments

- Big Data
- Randomised Controlled Trials
- Bias
- Observational studies

Data Scientist → solve problems with data + communicate

Ethics and Privacy

- Data provenance → documenting where Data comes from
- Data management → transparent plan for:
  1. Collection
  2. storage
  3. Exchange
  4. reporting & access.
- Data wrangling → transparent policy on how to deal w/ potential errors and missing values
- CARE principles (w/ Indigenous groups)
  - C – Collective benefits
  - A – Authority to control
  - R – Responsibility
  - E – Ethics

Big data

- Big data → massive amounts of data

RANDOMISED CONTROL TRIALS

what is link ?

(1) Domain knowledge → backgr info → data

RCT (Randomised Control Trial) → experiment designed to test whether a treatment causes an effect.

Types of evidence:

1. Personal testimony / Anecdotal evidence
2. Observational evidence
   → shows an association cause → effect
3. Experimental evidence
   → RCT

Bias → something that effects data to accurately measure treatment effect

→ SELECTION BIAS → some participants are more likely to be chosen than others.
→ e.g. Portacaval shunt 1966

Confounding / (confusion) → Treatment & Control group differ by some 3rd variable which influence the result being studied

→ "lurking variable" (hard to find)

→ Observer Bias → when participants or investigators are aware of the identity of the controlled & treatment group.

$\boxed{\text{Placebo effect}}$ → occurs when participant thinks they are having a treat

SOLUTION → DOUBLE BLIND RCT

→ neither particip. or investig. knows who is who

→ Random → treat
→ control

To achieve use a 3rd party to administer the treatment.

Biases

① Consent bias → when participants choose whether or not to take part in experim

② Survivor bias → people drop out of the experiment

③ Adherer bias → participants keep taking the placebo → as opposed to non-adherers.

Gold Standard

① RCT, double blind → minimises biases

Observational Studies

→ investigator CANNOT randomise groups
(NO RANDOMISATION → no direct correlation only association)

Precautions

① observational studies CANNOT establish CAUSATION
- Only establish an association

② Observ. studies MAY present as a RCT

i.e. a RCT is actually an observational study ∵ of confounding variables.

- control group → historical ≠ contemporaneous

For a RCT → must have a Contemporaneous Control Group → i.e. group occurs at same time as the treatment group.

③ Confounding variables can cause Simpson's Paradox

Simpson's Paradox: a trend in individual subgroups disappears or reverses when the groups are combined because of confounding variables

Dealing w/ Confounders:

① Randomisation ② Matching ③ Stratification

④ Statistical Adjustment.

CONTROL

a control
↳ person w/no treatment

control experiment
↓
investigator allocate subjects to 2 groups

Controlling for confounds
⟶ reduce influence of Confounding variables